

MODULE 05 · DOMAIN 2 - DESIGN RESILIENT ARCHITECTURES

High Availability and Fault Tolerance

Availability Zones, Auto Scaling, load balancers and DNS.

DOMAIN 2 of 4 EXAM WEIGHT 26% MODULE 5 of 22

Availability Zones are **physically separate**, and that is why 'across two AZs' is the most common right answer on this exam.

The levels of resilience, in order

- ◆ **Single instance.** Any failure is an outage. Only acceptable for development.
- ◆ **Multiple instances, one AZ.** Survives an instance failure, not a zone failure.
- ◆ **Multiple instances across AZs.** The default expectation for production. Survives losing a data centre.
- ◆ **Multiple Regions.** Survives a Region failure. Much more complex and expensive. Only when the question demands it.
- ◆ **Multi-Region active-active.** Highest availability and highest cost. Needs a specific justification in the question.

ON THE EXAM

Do not jump to multi-Region. It is correct only when the question names a Region-wide failure or a global latency requirement.

Choosing a load balancer

LOAD BALANCER	OPERATES AT	CHOOSE IT WHEN
Application Load Balancer	Layer 7, HTTP and HTTPS	Routing by path or host, containers, most web traffic
Network Load Balancer	Layer 4, TCP and UDP	Extreme throughput, static IP needed, or non-HTTP protocols
Gateway Load Balancer	Layer 3, appliances	Inserting third-party firewalls or inspection appliances

ON THE EXAM

Static IP address or a protocol that is not HTTP means Network Load Balancer.
Everything else web-shaped means Application Load Balancer.

How Auto Scaling actually decides

01

Launch template

Defines what an instance looks like when created.

02

Min, max, desired

The boundaries. Desired is what it aims for right now.

03

Scaling policy

Target tracking, step, or scheduled. Target tracking is the usual answer.

04

Health check

EC2 status, or ELB health check for application-level failure.

05

Replace

An unhealthy instance is terminated and a new one launched.

ON THE EXAM

Use the ELB health check type when the instance can be running but the application is broken.

Scaling policies, and when each is right

- ◆ **Target tracking.** Maintain a metric at a value. Simplest and the usual correct answer.
- ◆ **Step scaling.** Different responses at different alarm thresholds. For finer control.
- ◆ **Scheduled scaling.** For load that is predictable by the clock or calendar.
- ◆ **Predictive scaling.** Uses historical patterns to scale ahead of demand, not after it.
- ◆ **The warm-up problem.** If instances take minutes to become useful, reactive scaling arrives late.

ON THE EXAM

Known busy period every Monday morning means scheduled scaling. Recurring but discovered from history means predictive.

Route 53 routing policies

POLICY	WHAT IT DOES
Simple	One record, one answer. No health checks.
Failover	Primary until its health check fails, then secondary. Active-passive.
Weighted	Splits traffic by assigned proportion. Canary releases and gradual migration.
Latency	Sends the user to the Region with the lowest latency for them.
Geolocation	Routes by where the user is. For compliance or localised content.
Geoproximity	Routes by distance, with a bias you can adjust to shift traffic.
Multivalue answer	Returns several healthy records. Basic distribution, not a load balancer.

ON THE EXAM

Failover is active-passive DR. Latency is performance. Geolocation is compliance. These three are the most confused.

Route 53 latency routing or Global Accelerator?

ROUTE 53 LATENCY ROUTING

Chooses at DNS resolution time

- Free apart from normal query charges
- Subject to DNS caching and TTL
- Failover takes as long as the TTL
- Works with any endpoint

AWS GLOBAL ACCELERATOR

Chooses on the network path

- Two static anycast IP addresses
- Traffic enters the AWS backbone at the nearest edge
- Failover in seconds, independent of DNS
- Hourly charge plus data transfer

THE TIEBREAKER

Static IPs, non-HTTP protocols, or failover faster than DNS can propagate mean Global Accelerator.

A web application runs on four EC2 instances in a single Availability Zone behind an Application Load Balancer, with an Amazon RDS for MySQL instance in the same zone.

During a zone impairment last month the application was unavailable for six hours. The business wants to survive the loss of one Availability Zone without changing the application code.

Which combination of changes achieves this? (Choose two.)

- ☐ A Spread the instances across two Availability Zones in an Auto Scaling group
- ☐ B Convert the RDS instance to a Multi-AZ deployment
- ☐ C Create an RDS read replica in the same Availability Zone
- ☐ D Increase the instance size so fewer instances are needed

◆ PAUSE AND DECIDE

Which combination of changes achieves this? (Choose two.)

- A

 Instances across two AZs
Removes the compute single point of failure. The load balancer keeps serving from the surviving zone.
- B

 RDS Multi-AZ
Maintains a synchronous standby in another zone with automatic failover, and the endpoint name does not change, so no code changes.
- C

 Read replica in the same AZ
Same zone means the same failure takes both. Read replicas are also for scaling reads, not automatic failover.
- D

 Larger instances
Fewer, bigger instances in one zone is the same single point of failure with less redundancy.

THE SIGNAL

Multi-AZ is for availability. Read replicas are for read scaling. Confusing the two is the single most common RDS error.

An Auto Scaling group maintains six instances behind an Application Load Balancer. Monitoring shows all six are marked healthy, yet roughly one in six user requests returns a 502 error.

Investigation finds that on one instance the application process crashed three days ago. The operating system is running normally and the instance responds to pings.

What change would cause Auto Scaling to replace that instance?

- ☐ A Change the Auto Scaling group's health check type to ELB
- ☐ B Reduce the health check grace period to thirty seconds
- ☐ C Increase the desired capacity so healthy instances outnumber the failure
- ☐ D Enable detailed CloudWatch monitoring on the instances

◆ PAUSE AND DECIDE

What change would cause Auto Scaling to replace that instance?

- A

Health check type to ELB
The load balancer tests the application itself, so a crashed process fails the check and Auto Scaling replaces the instance.
- B

Shorter grace period
Controls how long a new instance is given to start before checks apply. Irrelevant to an instance running for three days.
- C

More instances
Dilutes the error rate without fixing anything. The broken instance stays in rotation indefinitely.
- D

Detailed monitoring
Improves metric resolution. It does not change what Auto Scaling considers healthy.

THE SIGNAL

'Instance is running but the application is not responding' is always the EC2 versus ELB health check distinction.

A multiplayer game server uses a custom protocol over UDP. Players connect to an endpoint whose address must be added to corporate firewall allow-lists, so it cannot change.

The company runs servers in three Regions and needs players routed to the nearest healthy Region, with failover in a few seconds if a Region becomes unavailable.

Which solution meets these requirements?

- ☐ A Route 53 latency-based routing to an Application Load Balancer in each Region
- ☐ B AWS Global Accelerator with endpoint groups in the three Regions
- ☐ C Amazon CloudFront with the game servers as custom origins
- ☐ D Route 53 failover routing with a short TTL

◆ PAUSE AND DECIDE

Which solution meets these requirements?

- A** Route 53 latency to ALBs
Two failures: an ALB cannot handle UDP, and DNS gives no static address and no seconds-level failover.
- B** Global Accelerator
Provides two static anycast IPs, supports TCP and UDP, routes over the AWS backbone to the nearest healthy Region, and fails over in seconds.
- C** CloudFront custom origins
CloudFront is an HTTP and HTTPS cache. It does not proxy a custom UDP protocol.
- D** Route 53 failover with short TTL
Active-passive rather than nearest, and clients frequently ignore short TTLs, so seconds-level failover is not achievable.

THE SIGNAL

Static IP plus UDP plus fast failover is the exact signature of Global Accelerator. Any one of the three should make you consider it.

An internal reporting system is used heavily between 08:00 and 10:00 each weekday and is almost idle otherwise. The instances take roughly seven minutes to boot and warm their caches.

With target tracking on CPU, users report that the first thirty minutes of each morning are slow, although the system performs well afterwards.

What is the MOST appropriate change?

- ☐ A Lower the target tracking CPU threshold so scaling triggers sooner
- ☐ B Add scheduled scaling to raise desired capacity before 08:00
- ☐ C Run the maximum capacity continuously to remove scaling delays
- ☐ D Replace the Auto Scaling group with a larger single instance

◆ PAUSE AND DECIDE

What is the MOST appropriate change?

A Lower the CPU threshold
Helps slightly but still reacts after load arrives, and seven-minute boots mean capacity still lands late. Also raises cost all day.

C Run at maximum always
Solves performance by paying for peak capacity twenty-four hours a day for a two-hour workload.

B Scheduled scaling before 08:00
The load is predictable by the clock, so capacity can be in place before demand rather than chasing it.

D One larger instance
Removes redundancy entirely and still has to be sized for peak.

THE SIGNAL

Predictable by the clock plus slow start-up means scheduled scaling. Reactive policies always arrive after the load.

A company is migrating a legacy application to AWS over six months. During the transition, both the on-premises system and the new AWS system must serve live traffic, with the proportion shifting gradually from one to the other.

The team wants to move ten percent of users at a time, monitor error rates, and roll back instantly if problems appear.

Which Route 53 routing policy supports this?

- ☐ A Failover routing
- ☐ B Weighted routing
- ☐ C Geolocation routing
- ☐ D Multivalue answer routing

◆ PAUSE AND DECIDE

Which Route 53 routing policy supports this?

A Failover

Active-passive. All traffic goes to the primary until it breaks. It cannot split traffic by proportion.

C Geolocation

Splits by where users are, not by proportion. It would migrate whole countries rather than a controlled percentage.

B Weighted

Assigns a proportion to each endpoint, so you can move ten percent at a time and revert by changing the weights back.

D Multivalue answer

Returns several healthy records for basic distribution, with no control over the split.

THE SIGNAL

Gradual migration, canary release, or A/B testing all mean weighted routing.

A stateless web tier runs across three Availability Zones in an Auto Scaling group. The team proposes adding a second Region in active-active configuration to improve availability, at roughly double the infrastructure cost.

The application has had two outages in two years. Both were caused by faulty deployments rather than AWS infrastructure failures.

What should the solutions architect advise?

- A Proceed with the second Region, since multi-Region is the highest availability design
- B Advise that a second Region would not have prevented either outage, and address the deployment process instead
- C Move to a fourth Availability Zone in the existing Region
- D Replace the Auto Scaling group with reserved capacity to prevent launch failures

◆ PAUSE AND DECIDE

What should the solutions architect advise?

- A

 Proceed with multi-Region
Doubles cost and complexity to protect against a failure mode that has not occurred, while leaving the actual cause untouched.
- B

 Fix the deployment process
A bad deployment replicates to both Regions, so multi-Region would not have prevented either outage. Match the remedy to the actual failure mode.
- C

 Fourth Availability Zone
Three zones already tolerate a zone failure. A fourth adds cost against a risk that is already handled.
- D

 Reserved capacity
Addresses capacity availability, which was not the cause of either outage.

THE SIGNAL

Match the remedy to the failure that actually occurred. The most impressive architecture is not the correct answer when it does not address the problem.

Module 5 takeaways

- ◆ **Across two or more AZs** is the default correct answer. Reach for multi-Region only when the question demands it.
- ◆ **Multi-AZ is availability. Read replicas are read scaling.** Never confuse them.
- ◆ **Running but not serving** means switch the health check type to ELB.
- ◆ **Static IP, UDP, or seconds-level failover** means Global Accelerator.
- ◆ **Predictable by the clock** means scheduled scaling, not a lower threshold.

ON THE EXAM

For every design, ask what happens when one component disappears. The answer should be 'not much'.